On conjectural series involving the hyperbolic cosine function

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Abstract

We give a modular-form proof of Conjecture (1) and of the second conjecture from the MathOverflow post *"Conjectural series involving the hyperbolic cosine function"*. `https://mathoverflow.net/questions/508960/conjectural-series-involving-the-hyperbolic-cosine-function` For Conjecture (1), we prove that

$$\sum_{n=1}^{\infty}\left(\frac{n^{2m}}{\cosh(n\pi)-1}-\frac{(2^{2m+1}-(-1)^{m(m+1)/2}2^{m+1}+4)n^{2m}}{\cosh(2n\pi)-1}+\frac{2^{2m+2}n^{2m}}{\cosh(4n\pi)-1}\right)$$

is equal to $1/12$ for $m=0$, to $1/(2\pi^2)$ for $m=1$, and to 0 for all $m>1$. We also prove

$$\sum_{n=1}^{\infty}\left(\frac{4}{(\cosh(n\pi)-1)^2}-\frac{55}{(\cosh(2n\pi)-1)^2}+\frac{16}{(\cosh(4n\pi)-1)^2}\right)=\frac{77-234/\pi}{72}.$$

1 Statement of the identities

For $m\in\mathbb{N}\cup\{0\}$, define

$$S_m:=\sum_{n=1}^{\infty}\left(\frac{n^{2m}}{\cosh(n\pi)-1}-\frac{c_m n^{2m}}{\cosh(2n\pi)-1}+\frac{2^{2m+2}n^{2m}}{\cosh(4n\pi)-1}\right),$$

where

$$c_m=2^{2m+1}-(-1)^{m(m+1)/2}2^{m+1}+4.$$

Thus

$$c_0=4,\qquad c_1=16,\qquad c_2=44,\qquad c_3=116.$$

The first four explicit instances are

$$\begin{aligned}
S_0&=\sum_{n=1}^{\infty}\left(\frac{1}{\cosh(n\pi)-1}-\frac{4}{\cosh(2n\pi)-1}+\frac{4}{\cosh(4n\pi)-1}\right),\\
S_1&=\sum_{n=1}^{\infty}\left(\frac{n^2}{\cosh(n\pi)-1}-\frac{16n^2}{\cosh(2n\pi)-1}+\frac{16n^2}{\cosh(4n\pi)-1}\right),\\
S_2&=\sum_{n=1}^{\infty}\left(\frac{n^4}{\cosh(n\pi)-1}-\frac{44n^4}{\cosh(2n\pi)-1}+\frac{64n^4}{\cosh(4n\pi)-1}\right),\\
S_3&=\sum_{n=1}^{\infty}\left(\frac{n^6}{\cosh(n\pi)-1}-\frac{116n^6}{\cosh(2n\pi)-1}+\frac{256n^6}{\cosh(4n\pi)-1}\right).
\end{aligned}$$

We first prove

$$S_0=\frac{1}{12},\qquad S_1=\frac{1}{2\pi^2},\qquad S_2=0,\qquad S_3=0,$$

and later we prove Conjecture (1) in full generality.

2 Definitions and standard identities

2.1 Eisenstein series

Let $\tau \in \mathfrak{H} := \{\tau \in \mathbb{C} : \Im\tau > 0\}$ and set

$$q = e^{2\pi i\tau}.$$

For $r \geq 0$, write

$$\sigma_r(n) := \sum_{d|n} d^r.$$

We write B_n for the Bernoulli numbers, defined by

$$\frac{t}{e^t - 1} = \sum_{n=0}^{\infty} B_n \frac{t^n}{n!}.$$

We use the quasimodular Eisenstein series

$$E_2(\tau) = 1 - 24 \sum_{n=1}^{\infty} \sigma_1(n) q^n,$$

and, for every even integer $k \geq 4$, the normalized Eisenstein series

$$E_k(\tau) = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n.$$

In particular,

$$E_4(\tau) = 1 + 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n,$$
$$E_6(\tau) = 1 - 504 \sum_{n=1}^{\infty} \sigma_5(n) q^n.$$

We also use the Ramanujan differential operator

$$D := q\frac{d}{dq} = \frac{1}{2\pi i}\frac{d}{d\tau}.$$

Ramanujan's differential identities are

$$DE_2 = \frac{E_2^2 - E_4}{12}, \tag{1}$$
$$DE_4 = \frac{E_2 E_4 - E_6}{3}, \tag{2}$$
$$DE_6 = \frac{E_2 E_6 - E_4^2}{2}. \tag{3}$$

For every even integer $k \geq 4$, define the classical Eisenstein series

$$G_k(\tau) := 2\zeta(k) E_k(\tau) = \sum_{(u,v)\in\mathbb{Z}^2\setminus\{(0,0)\}} (u + v\tau)^{-k}.$$

For $k \geq 4$, the defining series for G_k and its termwise derivative converge absolutely and locally uniformly on $\mathfrak{H}$. Moreover,

$$G_k\left(-\frac{1}{\tau}\right) = \tau^k G_k(\tau). \tag{4}$$

In particular,

$$E_4\left(-\frac{1}{\tau}\right) = \tau^4 E_4(\tau), \tag{5}$$

$$E_6\left(-\frac{1}{\tau}\right) = \tau^6 E_6(\tau). \tag{6}$$

For E_2, one has the quasimodular transformation law

$$E_2\left(-\frac{1}{\tau}\right) = \tau^2 E_2(\tau) + \frac{6\tau}{\pi i}. \tag{7}$$

2.2 Jacobi theta functions

Let

$$Q = e^{\pi i \tau}.$$

We use the classical theta functions

$$\vartheta_2(\tau) := \sum_{n\in\mathbb{Z}} Q^{(n+1/2)^2},$$
$$\vartheta_3(\tau) := \sum_{n\in\mathbb{Z}} Q^{n^2},$$
$$\vartheta_4(\tau) := \sum_{n\in\mathbb{Z}} (-1)^n Q^{n^2}.$$

The identities we need are:

$$\vartheta_3(\tau)^4 = \vartheta_2(\tau)^4 + \vartheta_4(\tau)^4, \tag{8}$$

$$\vartheta_2(\tau)^2 = 2\vartheta_2(2\tau)\vartheta_3(2\tau), \tag{9}$$

$$\vartheta_3(\tau)^2 = \vartheta_3(2\tau)^2 + \vartheta_2(2\tau)^2, \tag{10}$$

$$\vartheta_4(\tau)^2 = \vartheta_3(2\tau)^2 - \vartheta_2(2\tau)^2. \tag{11}$$

Moreover, under the inversion $\tau \mapsto -1/\tau$ one has

$$\vartheta_2\left(-\frac{1}{\tau}\right) = (-i\tau)^{1/2}\vartheta_4(\tau), \qquad \vartheta_4\left(-\frac{1}{\tau}\right) = (-i\tau)^{1/2}\vartheta_2(\tau).$$

At the fixed point $\tau = i$ this implies

$$\vartheta_2(i) = \vartheta_4(i). \tag{12}$$

We also need the standard formulas expressing Eisenstein series in theta constants:

$$2E_2(2\tau) - E_2(\tau) = \vartheta_3(2\tau)^4 + \vartheta_2(2\tau)^4, \tag{13}$$

$$E_4(\tau) = \frac{1}{2}\big(\vartheta_2(\tau)^8 + \vartheta_3(\tau)^8 + \vartheta_4(\tau)^8\big), \tag{14}$$

$$E_6(\tau) = \frac{1}{2}\big(\vartheta_2(\tau)^4 + \vartheta_3(\tau)^4\big)\big(\vartheta_3(\tau)^4 + \vartheta_4(\tau)^4\big)\big(\vartheta_4(\tau)^4 - \vartheta_2(\tau)^4\big). \tag{15}$$

3 Reduction of the hyperbolic sums to Eisenstein series

Define, for $a > 0$ and $m \geq 0$,

$$L_m(a) := \sum_{n=1}^{\infty} \frac{n^{2m}}{\cosh(a\pi n) - 1}.$$

Then

$$S_m = L_m(1) - c_m L_m(2) + 2^{2m+2} L_m(4).$$

Lemma 1. *For every $x > 0$,*

$$\frac{1}{\cosh x - 1} = \frac{2e^{-x}}{(1 - e^{-x})^2} = 2 \sum_{r=1}^{\infty} r e^{-rx}.$$

Consequently, if $\tau = ia/2$ (so that $q = e^{-a\pi}$), then

$$L_m(a) = 2 \sum_{N=1}^{\infty} N \sigma_{2m-1}(N) q^N.$$

For $m = 0$,

$$L_0(a) = \frac{1 - E_2(\tau)}{12}. \tag{16}$$

For every $m \geq 1$,

$$L_m(a) = -\frac{B_{2m}}{2m} DE_{2m}(\tau). \tag{17}$$

In particular,

$$L_1(a) = \frac{E_4(\tau) - E_2(\tau)^2}{144}, \tag{18}$$

$$L_2(a) = \frac{E_2(\tau)E_4(\tau) - E_6(\tau)}{360}, \tag{19}$$

$$L_3(a) = \frac{E_4(\tau)^2 - E_2(\tau)E_6(\tau)}{504}. \tag{20}$$

Proof. The expansion of $1/(\cosh x - 1)$ is immediate:

$$\cosh x - 1 = \frac{e^x + e^{-x} - 2}{2} = \frac{(1 - e^{-x})^2}{2e^{-x}},$$

so

$$\frac{1}{\cosh x - 1} = \frac{2e^{-x}}{(1 - e^{-x})^2} = 2 \sum_{r=1}^{\infty} r e^{-rx}.$$

Therefore

$$L_m(a) = 2 \sum_{n=1}^{\infty} \sum_{r=1}^{\infty} r\, n^{2m} q^{nr}.$$

Collecting the coefficient of q^N gives

$$\sum_{nr=N} r n^{2m} = \sum_{n|N} \frac{N}{n} n^{2m} = N \sum_{n|N} n^{2m-1} = N \sigma_{2m-1}(N),$$

which proves

$$L_m(a) = 2\sum_{N=1}^{\infty} N\sigma_{2m-1}(N)q^N.$$

For $m = 0$, this gives

$$L_0(a) = 2\sum_{N=1}^{\infty} \sigma_1(N)q^N = \frac{1 - E_2(\tau)}{12}.$$

Now let $m \geq 1$. From the definition of E_{2m},

$$E_{2m}(\tau) = 1 - \frac{4m}{B_{2m}}\sum_{N=1}^{\infty} \sigma_{2m-1}(N)q^N,$$

hence

$$DE_{2m}(\tau) = -\frac{4m}{B_{2m}}\sum_{N=1}^{\infty} N\sigma_{2m-1}(N)q^N = -\frac{2m}{B_{2m}}L_m(a).$$

This proves (17).

For $m = 1$, (17) gives

$$L_1(a) = -\frac{1}{12}DE_2(\tau),$$

and (1) yields (18). For $m = 2$, one has

$$L_2(a) = \frac{1}{120}DE_4(\tau),$$

and (2) yields (19). For $m = 3$, one has

$$L_3(a) = -\frac{1}{252}DE_6(\tau),$$

and (3) yields (20). □

4 Special identities at $\tau = i$ and $\tau = 2i$

Set

$$a := \vartheta_3(2i), \qquad b := \vartheta_2(2i), \qquad A := a^4 = \vartheta_3(2i)^4, \qquad B := b^4 = \vartheta_2(2i)^4.$$

Lemma 2. *The numbers A and B satisfy*

$$A^2 - 34AB + B^2 = 0. \tag{21}$$

Proof. By (9) and (11), evaluated at $\tau = i$,

$$\vartheta_2(i)^2 = 2ab, \qquad \vartheta_4(i)^2 = a^2 - b^2.$$

Using (12), we get

$$2ab = a^2 - b^2.$$

Since $a, b > 0$, dividing by b^2 yields

$$\left(\frac{a}{b}\right)^2 - 2\frac{a}{b} - 1 = 0.$$

Thus

$$\frac{a}{b} = 1 + \sqrt{2}.$$

Raising to the fourth power gives

$$\frac{A}{B} = \left(1 + \sqrt{2}\right)^4 = 17 + 12\sqrt{2}.$$

Since $(17 + 12\sqrt{2}) + (17 - 12\sqrt{2}) = 34$ and $(17 + 12\sqrt{2})(17 - 12\sqrt{2}) = 1$, the ratio A/B satisfies

$$\left(\frac{A}{B}\right)^2 - 34\left(\frac{A}{B}\right) + 1 = 0.$$

Multiplying by B^2 gives (21). □

Lemma 3. *With the notation above, one has*

$$E_2(i) = \frac{3}{\pi}, \tag{22}$$

$$E_2(2i) = \frac{A+B}{2} + \frac{3}{2\pi}, \tag{23}$$

$$E_2(i/2) = -2(A+B) + \frac{6}{\pi}, \tag{24}$$

$$E_4(2i) = A^2 - AB + B^2, \tag{25}$$

$$E_4(i) = 48AB, \tag{26}$$

$$E_4(i/2) = 16E_4(2i), \tag{27}$$

$$E_6(i) = 0, \tag{28}$$

$$E_6(2i) = \frac{1}{2}(A+B)(2A-B)(A-2B), \tag{29}$$

$$E_6(i/2) = -64E_6(2i). \tag{30}$$

Proof. Applying (7) with $\tau = i$ gives

$$E_2(i) = i^2 E_2(i) + \frac{6i}{\pi i} = -E_2(i) + \frac{6}{\pi},$$

so (22) follows.

Next, (13) with $\tau = i$ gives

$$2E_2(2i) - E_2(i) = \vartheta_3(2i)^4 + \vartheta_2(2i)^4 = A + B.$$

Using (22) we obtain (23). Then (7) with $\tau = 2i$ gives

$$E_2(i/2) = E_2\left(-\frac{1}{2i}\right) = (2i)^2 E_2(2i) + \frac{6(2i)}{\pi i} = -4E_2(2i) + \frac{12}{\pi},$$

which, combined with (23), yields (24).

For (25), use (14) at $\tau = 2i$ and Jacobi's identity (8):

$$\vartheta_4(2i)^4 = \vartheta_3(2i)^4 - \vartheta_2(2i)^4 = A - B.$$

Hence

$$E_4(2i) = \frac{1}{2}\left(B^2 + A^2 + (A-B)^2\right) = A^2 - AB + B^2.$$

To compute $E_4(i)$, note from (12) and (8) that

$$\vartheta_3(i)^4 = \vartheta_2(i)^4 + \vartheta_4(i)^4 = 2\vartheta_2(i)^4.$$

Therefore, by (14),

$$E_4(i) = \frac{1}{2}(\vartheta_2(i)^8 + \vartheta_3(i)^8 + \vartheta_4(i)^8) = \frac{1}{2}(x^2 + (2x)^2 + x^2) = 3x^2,$$

where $x := \vartheta_2(i)^4$. From (9),

$$x = \vartheta_2(i)^4 = (2ab)^2 = 4a^2b^2,$$

so

$$x^2 = 16a^4b^4 = 16AB.$$

Thus (26) follows:

$$E_4(i) = 3 \cdot 16AB = 48AB.$$

Equation (27) is immediate from (5) with $\tau = 2i$, because $i/2 = -1/(2i)$:

$$E_4(i/2) = (2i)^4 E_4(2i) = 16E_4(2i).$$

For (28), apply (6) with $\tau = i$:

$$E_6(i) = i^6 E_6(i) = -E_6(i),$$

so $E_6(i) = 0$.

For (29), use (15) at $\tau = 2i$. Since

$$\vartheta_2(2i)^4 = B, \qquad \vartheta_3(2i)^4 = A, \qquad \vartheta_4(2i)^4 = A - B,$$

we get

$$E_6(2i) = \frac{1}{2}(A+B)(A+A-B)(A-B-B) = \frac{1}{2}(A+B)(2A-B)(A-2B).$$

Finally, (6) with $\tau = 2i$ gives

$$E_6(i/2) = E_6\left(-\frac{1}{2i}\right) = (2i)^6 E_6(2i) = -64E_6(2i),$$

which is (30). □

5 Initial cases and subsequent generalizations

Theorem 1. *The sums S_0, S_1, S_2, S_3 satisfy*

$$S_0 = \frac{1}{12}, \qquad S_1 = \frac{1}{2\pi^2}, \qquad S_2 = 0, \qquad S_3 = 0.$$

Proof. We treat the four cases separately.

Case $m = 0$. By (16),

$$S_0 = L_0(1) - 4L_0(2) + 4L_0(4) = \frac{1 - E_2(i/2) + 4E_2(i) - 4E_2(2i)}{12}.$$

Substituting (22) and (24), we get

$$\begin{aligned} 12S_0 &= 1 - \left(-4E_2(2i) + \frac{12}{\pi}\right) + 4 \cdot \frac{3}{\pi} - 4E_2(2i) \\ &= 1. \end{aligned}$$

Hence

$$S_0 = \frac{1}{12}.$$

Case $m = 1$. By (18),

$$144S_1 = (E_4 - E_2^2)(i/2) - 16(E_4 - E_2^2)(i) + 16(E_4 - E_2^2)(2i).$$

Equivalently,

$$144S_1 = E_4(i/2) - 16E_4(i) + 16E_4(2i) - E_2(i/2)^2 + 16E_2(i)^2 - 16E_2(2i)^2.$$

Using (27), this becomes

$$144S_1 = 32E_4(2i) - 16E_4(i) - E_2(i/2)^2 + 16E_2(i)^2 - 16E_2(2i)^2.$$

Now substitute (22), (23), (24), (25), and (26):

$$E_2(i) = \frac{3}{\pi}, \qquad E_2(2i) = \frac{A+B}{2} + \frac{3}{2\pi}, \qquad E_2(i/2) = -2(A+B) + \frac{6}{\pi},$$

$$E_4(2i) = A^2 - AB + B^2, \qquad E_4(i) = 48AB.$$

A direct expansion gives

$$144S_1 = 24A^2 - 816AB + 24B^2 + \frac{72}{\pi^2}.$$

Factor out 24 and use (21):

$$144S_1 = 24(A^2 - 34AB + B^2) + \frac{72}{\pi^2} = \frac{72}{\pi^2}.$$

Therefore

$$S_1 = \frac{1}{2\pi^2}.$$

Case $m = 2$. By (19),

$$360S_2 = (E_2E_4 - E_6)(i/2) - 44(E_2E_4 - E_6)(i) + 64(E_2E_4 - E_6)(2i).$$

Since $E_6(i) = 0$, $E_4(i/2) = 16E_4(2i)$, and $E_6(i/2) = -64E_6(2i)$, the E_6-terms cancel and we obtain

$$\begin{aligned} 360S_2 &= E_2(i/2)E_4(i/2) - 44E_2(i)E_4(i) + 64E_2(2i)E_4(2i) \\ &= 16E_4(2i)\big(E_2(i/2) + 4E_2(2i)\big) - 44E_2(i)E_4(i). \end{aligned}$$

By (22), (24), and (23),

$$E_2(i/2) + 4E_2(2i) = \frac{12}{\pi}.$$

Hence

$$360S_2 = \frac{12}{\pi}\big(16E_4(2i) - 11E_4(i)\big).$$

Using (25) and (26),

$$16E_4(2i) - 11E_4(i) = 16(A^2 - AB + B^2) - 11 \cdot 48AB = 16(A^2 - 34AB + B^2) = 0$$

by (21). Therefore

$$S_2 = 0.$$

Case $m = 3$. By (20),

$$504S_3 = (E_4^2 - E_2E_6)(i/2) - 116(E_4^2 - E_2E_6)(i) + 256(E_4^2 - E_2E_6)(2i).$$

Using $E_6(i) = 0$, $E_4(i/2) = 16E_4(2i)$, and $E_6(i/2) = -64E_6(2i)$, we get

$$\begin{aligned} 504S_3 &= 256E_4(2i)^2 - 116E_4(i)^2 + 64E_2(i/2)E_6(2i) + 256E_4(2i)^2 - 256E_2(2i)E_6(2i) \\ &= 512E_4(2i)^2 - 116E_4(i)^2 + 64\big(E_2(i/2) - 4E_2(2i)\big)E_6(2i). \end{aligned}$$

By (24) and (23),

$$E_2(i/2) - 4E_2(2i) = -4(A + B).$$

Substituting (25), (26), and (29), we obtain

$$\begin{aligned} 504S_3 = 512(A^2 - AB + B^2)^2 - 116(48AB)^2 \\ - 128(A + B)^2(2A - B)(A - 2B). \end{aligned}$$

A straightforward expansion and factorization yields

$$504S_3 = 128\big(A^2 - 34AB + B^2\big)\big(2A^2 + 61AB + 2B^2\big).$$

By (21), the first factor vanishes. Hence

$$S_3 = 0.$$

This completes the proof. □

6 Final remark

The same mechanism explains why these identities are naturally related to modular and quasimodular forms of level 2: after rewriting

$$\frac{1}{\cosh x - 1} = 2\sum_{r=1}^{\infty} re^{-rx},$$

the sums become linear combinations of values of DE_{2m} at the CM points $i/2$, i, and $2i$. The cases $m = 0, 1, 2, 3$ collapse because of the single algebraic relation (21) among the theta-constants at $2i$.

The second conjecture

We now prove the second conjecture from the MathOverflow post, namely

$$\sum_{n=1}^{\infty}\left(\frac{4}{(\cosh(n\pi) - 1)^2} - \frac{55}{(\cosh(2n\pi) - 1)^2} + \frac{16}{(\cosh(4n\pi) - 1)^2}\right) = \frac{77 - 234/\pi}{72}.$$

Theorem 2. *One has*

$$\sum_{n=1}^{\infty}\left(\frac{4}{(\cosh(n\pi) - 1)^2} - \frac{55}{(\cosh(2n\pi) - 1)^2} + \frac{16}{(\cosh(4n\pi) - 1)^2}\right) = \frac{77 - 234/\pi}{72}.$$

Proof. For $a > 0$, define

$$T(a) := \sum_{n=1}^{\infty} \frac{1}{(\cosh(a\pi n) - 1)^2}.$$

Then the required identity is

$$4T(1) - 55T(2) + 16T(4) = \frac{77 - 234/\pi}{72}.$$

We first express $T(a)$ in terms of Eisenstein series. Since

$$\cosh x - 1 = \frac{(1 - e^{-x})^2}{2e^{-x}},$$

we have

$$\frac{1}{(\cosh x - 1)^2} = \frac{4e^{-2x}}{(1 - e^{-x})^4}.$$

Now

$$\frac{1}{(1-y)^4} = \sum_{r=0}^{\infty} \binom{r+3}{3} y^r,$$

so with $y = e^{-x}$,

$$\frac{4e^{-2x}}{(1 - e^{-x})^4} = 4 \sum_{r=0}^{\infty} \binom{r+3}{3} e^{-(r+2)x}.$$

Writing $m = r + 2$, we obtain

$$\frac{1}{(\cosh x - 1)^2} = 4 \sum_{m=2}^{\infty} \binom{m+1}{3} e^{-mx} = \frac{2}{3} \sum_{m=1}^{\infty} (m^3 - m) e^{-mx}.$$

Therefore, if

$$q = e^{-a\pi} = e^{2\pi i \tau}, \qquad \tau = \frac{ia}{2},$$

then

$$T(a) = \frac{2}{3} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (m^3 - m) q^{mn}.$$

Collecting the coefficient of q^N gives

$$\sum_{m|N} (m^3 - m) = \sigma_3(N) - \sigma_1(N),$$

hence

$$T(a) = \frac{2}{3} \sum_{N=1}^{\infty} (\sigma_3(N) - \sigma_1(N)) q^N.$$

Using

$$E_2(\tau) = 1 - 24 \sum_{N=1}^{\infty} \sigma_1(N) q^N, \qquad E_4(\tau) = 1 + 240 \sum_{N=1}^{\infty} \sigma_3(N) q^N,$$

we get

$$\sum_{N=1}^{\infty} \sigma_1(N) q^N = \frac{1 - E_2(\tau)}{24}, \qquad \sum_{N=1}^{\infty} \sigma_3(N) q^N = \frac{E_4(\tau) - 1}{240}.$$

Therefore

$$T(a) = \frac{2}{3}\left(\frac{E_4(\tau)-1}{240} - \frac{1-E_2(\tau)}{24}\right) = \frac{E_4(\tau)+10E_2(\tau)-11}{360}.$$

Now define

$$C := 4T(1) - 55T(2) + 16T(4).$$

Since $a = 1, 2, 4$ correspond to $\tau = i/2, i, 2i$, we obtain

$$360C = 4\big(E_4(i/2) + 10E_2(i/2) - 11\big) - 55\big(E_4(i) + 10E_2(i) - 11\big) + 16\big(E_4(2i) + 10E_2(2i) - 11\big).$$

Expanding,

$$360C = \big(4E_4(i/2) - 55E_4(i) + 16E_4(2i)\big) + \big(40E_2(i/2) - 550E_2(i) + 160E_2(2i)\big) + 385.$$

We now simplify the E_2-part. By (22), (23), and (24),

$$E_2(i) = \frac{3}{\pi}, \qquad E_2(i/2) + 4E_2(2i) = \frac{12}{\pi}.$$

Hence

$$40E_2(i/2) - 550E_2(i) + 160E_2(2i) = 40\big(E_2(i/2) + 4E_2(2i)\big) - 550E_2(i) = 40 \cdot \frac{12}{\pi} - 550 \cdot \frac{3}{\pi} = -\frac{1170}{\pi}.$$

Next we simplify the E_4-part. By (27),

$$E_4(i/2) = 16E_4(2i),$$

so

$$4E_4(i/2) - 55E_4(i) + 16E_4(2i) = 80E_4(2i) - 55E_4(i) = 5\big(16E_4(2i) - 11E_4(i)\big).$$

Using (25), (26), and (21), we obtain

$$16E_4(2i) - 11E_4(i) = 16(A^2 - AB + B^2) - 11 \cdot 48AB = 16(A^2 - 34AB + B^2) = 0.$$

Therefore the entire E_4-part vanishes.

Substituting back, we find

$$360C = 385 - \frac{1170}{\pi}.$$

Thus

$$C = \frac{385 - 1170/\pi}{360} = \frac{77 - 234/\pi}{72}.$$

Since $C = 4T(1) - 55T(2) + 16T(4)$, this is exactly

$$\sum_{n=1}^{\infty}\left(\frac{4}{(\cosh(n\pi)-1)^2} - \frac{55}{(\cosh(2n\pi)-1)^2} + \frac{16}{(\cosh(4n\pi)-1)^2}\right) = \frac{77 - 234/\pi}{72}.$$

This completes the proof. □

Conjecture (1) in full generality

We now prove Conjecture (1) for all $m \geq 0$. Recall that

$$S_m = \sum_{n=1}^{\infty} \left(\frac{n^{2m}}{\cosh(n\pi) - 1} - \frac{c_m n^{2m}}{\cosh(2n\pi) - 1} + \frac{2^{2m+2} n^{2m}}{\cosh(4n\pi) - 1} \right),$$

where

$$c_m = 2^{2m+1} - (-1)^{m(m+1)/2} 2^{m+1} + 4.$$

Theorem 3. *For all $m \geq 0$, one has*

$$S_0 = \frac{1}{12}, \qquad S_1 = \frac{1}{2\pi^2}, \qquad S_m = 0 \text{ for all } m > 1.$$

Proof. The cases $m = 0$ and $m = 1$ were proved above, so it remains to show that $S_m = 0$ for every $m > 1$.

Let

$$k := 2m, \qquad \varepsilon_m := (-1)^{m(m+1)/2}.$$

Then $k \geq 4$. By Lemma 1, for $\tau = ia/2$,

$$L_m(a) = -\frac{B_k}{k} DE_k(\tau), \qquad D = \frac{1}{2\pi i} \frac{d}{d\tau}.$$

Hence

$$S_m = 0$$

is equivalent to

$$DE_k(i/2) - c_m DE_k(i) + 2^{k+2} DE_k(2i) = 0.$$

Since $D = \frac{1}{2\pi i} \frac{d}{d\tau}$, this is equivalent to

$$E_k'(i/2) - c_m E_k'(i) + 2^{k+2} E_k'(2i) = 0.$$

It is convenient to replace E_k by the classical Eisenstein series

$$G_k(\tau) := 2\zeta(k) E_k(\tau) = \sum_{(u,v) \in \mathbb{Z}^2 \setminus \{(0,0)\}} (u + v\tau)^{-k}.$$

Since $2\zeta(k) \neq 0$, it is enough to prove

$$G_k'(i/2) - c_m G_k'(i) + 2^{k+2} G_k'(2i) = 0. \tag{31}$$

For $k \geq 4$, the defining series for G_k and G_k' converge absolutely and locally uniformly on $\mathfrak{H}$, so all rearrangements and termwise differentiations below are justified.

Step 1: parity class sums. For $\alpha, \beta \in \{0, 1\}$, define

$$\Sigma_{\alpha,\beta}(\tau) := \sum_{r,s \in \mathbb{Z}}^{*_{\alpha,\beta}} (2r + \alpha + (2s + \beta)\tau)^{-k},$$

where the only omitted term is $r = s = 0$ when $(\alpha, \beta) = (0, 0)$. Then

$$G_k(\tau) = \Sigma_{00}(\tau) + \Sigma_{10}(\tau) + \Sigma_{01}(\tau) + \Sigma_{11}(\tau),$$

because every lattice point $(u, v) \in \mathbb{Z}^2$ falls into exactly one parity class modulo 2. Moreover,

$$G_k(2\tau) = \Sigma_{00}(\tau) + \Sigma_{10}(\tau),$$

because $G_k(2\tau)$ is the subsum over lattice points whose coefficient of τ is even, that is, over the classes with $\beta = 0$.

We abbreviate

$$\Sigma_{\alpha,\beta} := \Sigma_{\alpha,\beta}(i), \qquad \Sigma'_{\alpha,\beta} := \frac{d}{d\tau}\Sigma_{\alpha,\beta}(\tau)\Big|_{\tau=i}.$$

The class $(0, 0)$ is just a dilation, so

$$\Sigma_{00}(\tau) = 2^{-k}G_k(\tau), \qquad \Sigma'_{00} = 2^{-k}G'_k(i). \tag{32}$$

The classes with equal parity satisfy

$$\Sigma_{00} + \Sigma_{11} = \sideset{}{'}\sum_{\substack{u,v\in\mathbb{Z} \\ u\equiv v \pmod 2}} (u + vi)^{-k}.$$

The change of variables

$$u = a - b, \qquad v = a + b$$

is a bijection from $\mathbb{Z}^2$ onto the set of pairs $(u, v) \in \mathbb{Z}^2$ with $u \equiv v \pmod 2$. Under this change of variables,

$$u + vi = (a - b) + (a + b)i = (1 + i)(a + bi),$$

hence

$$\Sigma_{00} + \Sigma_{11} = (1 + i)^{-k}G_k(i). \tag{33}$$

Also, multiplication by i sends

$$2r + 1 + 2si \longmapsto i(2r + 1 + 2si) = -2s + (2r + 1)i,$$

which is exactly the class $(0, 1)$. Therefore

$$\Sigma_{01} = i^{-k}\Sigma_{10} = (-1)^m\Sigma_{10}. \tag{34}$$

Step 2: the case m even. Assume first that m is even. Then $k \equiv 0 \pmod 4$, so $(-1)^m = 1$, and (34) gives

$$\Sigma_{01} = \Sigma_{10}.$$

Therefore

$$G_k(i) = \Sigma_{00} + 2\Sigma_{10} + \Sigma_{11}.$$

Since $G_k(2i) = \Sigma_{00} + \Sigma_{10}$, we obtain

$$G_k(i) = 2G_k(2i) - \Sigma_{00} + \Sigma_{11}.$$

Using (32) and (33),

$$\Sigma_{11} = (1 + i)^{-k}G_k(i) - 2^{-k}G_k(i).$$

Hence

$$G_k(i) = 2G_k(2i) + \big((1 + i)^{-k} - 2^{1-k}\big)G_k(i).$$

Now

$$(1+i)^{-k} = 2^{-m}(-i)^m.$$

Since m is even, $(-i)^m = \varepsilon_m$. Thus

$$(1 - \varepsilon_m 2^{-m} + 2^{1-k})G_k(i) = 2G_k(2i).$$

Multiplying by 2^{k+1}, and recalling that $k = 2m$, yields

$$c_m\, G_k(i) = 2^{k+2} G_k(2i). \tag{35}$$

We now turn to derivatives. The modular transformation

$$G_k\left(-\frac{1}{\tau}\right) = \tau^k G_k(\tau)$$

implies, by differentiation,

$$G_k'\left(-\frac{1}{\tau}\right) = \tau^{k+2} G_k'(\tau) + k\tau^{k+1} G_k(\tau). \tag{36}$$

Evaluating at $\tau = i$, and using $i^k = 1$ because $k \equiv 0 \pmod 4$, we get

$$G_k'(i) = \frac{ki}{2} G_k(i).$$

Evaluating (36) at $\tau = 2i$, and using $i/2 = -1/(2i)$, gives

$$G_k'(i/2) = (2i)^{k+2} G_k'(2i) + k(2i)^{k+1} G_k(2i).$$

Since $k \equiv 0 \pmod 4$,

$$(2i)^{k+2} = -2^{k+2}, \qquad (2i)^{k+1} = 2^{k+1} i.$$

Therefore

$$G_k'(i/2) + 2^{k+2} G_k'(2i) = k 2^{k+1} i\, G_k(2i).$$

Subtracting $c_m G_k'(i) = \frac{ki}{2} c_m G_k(i)$, we obtain

$$G_k'(i/2) - c_m G_k'(i) + 2^{k+2} G_k'(2i) = ki\left(2^{k+1} G_k(2i) - \frac{c_m}{2} G_k(i)\right).$$

By (35), the bracket vanishes. Thus (31) holds when m is even.

Step 3: the case m odd. Now assume that m is odd. Then $k \equiv 2 \pmod 4$. From modularity at the fixed point i,

$$G_k(i) = i^k G_k(i) = -G_k(i),$$

hence

$$G_k(i) = 0. \tag{37}$$

By (32), (33), and (34), this implies

$$\Sigma_{00} = 0, \qquad \Sigma_{11} = 0, \qquad \Sigma_{01} = -\Sigma_{10}.$$

Since $G_k(2i) = \Sigma_{00} + \Sigma_{10}$, we obtain

$$G_k(2i) = \Sigma_{10}. \tag{38}$$

Next we compute the derivatives of the class sums.

First, differentiating the identity $G_k(2\tau) = \Sigma_{00}(\tau) + \Sigma_{10}(\tau)$ at $\tau = i$, we get

$$2G_k'(2i) = \Sigma_{00}' + \Sigma_{10}'. \tag{39}$$

Using (32), this gives

$$\Sigma_{10}' = 2G_k'(2i) - 2^{-k}G_k'(i). \tag{40}$$

Second, consider

$$\Sigma_{00}' + \Sigma_{11}' = -k \sideset{}{'}\sum_{\substack{u,v\in\mathbb{Z} \\ u\equiv v \pmod 2}} v\,(u+vi)^{-k-1}.$$

Again write $u = a - b$, $v = a + b$, so $u + vi = (1 + i)(a + bi)$. Then

$$\Sigma_{00}' + \Sigma_{11}' = -k(1+i)^{-k-1} \sum_{(a,b)\neq(0,0)} (a+b)(a+bi)^{-k-1}.$$

Now

$$a + b = \frac{1-i}{2}(a+bi) + \frac{1+i}{2}(a-bi),$$

hence

$$\Sigma_{00}' + \Sigma_{11}' = -\frac{k(1+i)^{-k-1}}{2} \sum_{(a,b)\neq(0,0)} \Big((1-i)(a+bi)^{-k} + (1+i)(a-bi)(a+bi)^{-k-1}\Big).$$

Because $G_k(i) = 0$, the first sum vanishes, and therefore

$$\Sigma_{00}' + \Sigma_{11}' = -\frac{k(1+i)^{-k}}{2} \sum_{(a,b)\neq(0,0)} (a-bi)(a+bi)^{-k-1}.$$

Set

$$A := \sum_{(a,b)\neq(0,0)} a(a+bi)^{-k-1}, \qquad B := \sum_{(a,b)\neq(0,0)} b(a+bi)^{-k-1}.$$

Then the last sum is $A - iB$. Under the rotation $(a, b) \mapsto (-b, a)$, we have

$$A = \sum_{(a,b)\neq(0,0)} (-b)(-b+ai)^{-k-1} = i^{-k-1}(-1) \sum_{(a,b)\neq(0,0)} b(a+bi)^{-k-1}.$$

Since $k \equiv 2 \pmod 4$, one has $i^{-k-1} = i$, so

$$A = -iB.$$

Thus

$$A - iB = -2iB.$$

On the other hand,

$$G_k'(i) = -k \sum_{(a,b)\neq(0,0)} b(a+bi)^{-k-1} = -kB,$$

so $B = -G_k'(i)/k$. Therefore

$$\Sigma_{00}' + \Sigma_{11}' = -i(1+i)^{-k}G_k'(i).$$

Since

$$(1+i)^{-k} = 2^{-m}(-i)^m,$$

and m is odd, one checks that

$$-i(1+i)^{-k} = \varepsilon_m 2^{-m}.$$

Hence

$$\Sigma'_{00} + \Sigma'_{11} = \varepsilon_m 2^{-m} G'_k(i). \tag{41}$$

Using (32), this yields

$$\Sigma'_{11} = \big(\varepsilon_m 2^{-m} - 2^{-k}\big) G'_k(i). \tag{42}$$

Third, we relate Σ'_{01} and Σ'_{10}. Every element of the class $(0,1)$ is iz with z in the class $(1,0)$. Hence

$$\Sigma'_{01} = -k \sum_{r,s\in\mathbb{Z}} (2r+1)\big(i(2r+1+2si)\big)^{-k-1}.$$

Since $k \equiv 2 \pmod 4$, we have $i^{-k-1} = i$. Writing

$$z := 2r+1+2si, \qquad 2r+1 = z - 2si,$$

we obtain

$$\Sigma'_{01} = -ki \sum z^{-k} - 2k \sum s\, z^{-k-1}.$$

The first sum is Σ_{10}, and the second is exactly Σ'_{10}. Therefore

$$\Sigma'_{01} = \Sigma'_{10} - ki\,\Sigma_{10}. \tag{43}$$

Using (38), this becomes

$$\Sigma'_{01} = \Sigma'_{10} - ki\, G_k(2i).$$

Now combine everything:

$$G'_k(i) = \Sigma'_{00} + \Sigma'_{10} + \Sigma'_{01} + \Sigma'_{11}.$$

Substituting (40), (42), and (43), we find

$$\begin{aligned} G'_k(i) &= 2^{-k} G'_k(i) + \Sigma'_{10} + \big(\Sigma'_{10} - ki\, G_k(2i)\big) + \big(\varepsilon_m 2^{-m} - 2^{-k}\big) G'_k(i) \\ &= 2\Sigma'_{10} - ki\, G_k(2i) + \varepsilon_m 2^{-m} G'_k(i). \end{aligned}$$

Using (40),

$$G'_k(i) = 4G'_k(2i) - 2^{1-k} G'_k(i) - ki\, G_k(2i) + \varepsilon_m 2^{-m} G'_k(i).$$

Hence

$$\big(1 - \varepsilon_m 2^{-m} + 2^{1-k}\big) G'_k(i) = 4G'_k(2i) - ki\, G_k(2i).$$

Multiplying by 2^{k+1} gives

$$c_m G'_k(i) = 2^{k+3} G'_k(2i) - k 2^{k+1} i\, G_k(2i). \tag{44}$$

Finally, apply (36) at $\tau = 2i$:

$$G'_k(i/2) = (2i)^{k+2} G'_k(2i) + k(2i)^{k+1} G_k(2i).$$

Since $k \equiv 2 \pmod 4$,

$$(2i)^{k+2} = 2^{k+2}, \qquad (2i)^{k+1} = -2^{k+1} i.$$

Therefore

$$G_k'(i/2) = 2^{k+2} G_k'(2i) - k 2^{k+1} i \, G_k(2i).$$

Subtracting $c_m G_k'(i)$ and adding $2^{k+2} G_k'(2i)$, and then using (44), we get

$$G_k'(i/2) - c_m G_k'(i) + 2^{k+2} G_k'(2i) = 0.$$

Thus (31) holds when m is odd as well.

We have therefore proved (31) for every $m > 1$, hence $S_m = 0$ for all $m > 1$. Together with the already established cases $m = 0$ and $m = 1$, this proves the theorem. $\square$